Skills Summary

Sorting Quadrilaterals by Their Attributes

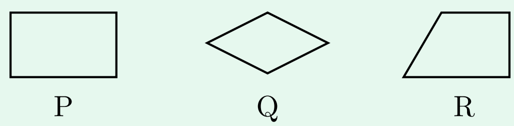
Use this reference while completing your worksheet or when studying.

Skill 1 — Checking One Shape's Attributes

A quadrilateral is any closed shape with four straight sides. Three attributes tell them apart:

square corners — corners that the corner of a card fits exactly;
parallel sides — two sides that stay the same distance apart all the way along;
equal sides — sides of the same length. Check each attribute separately.

Example: How many square corners does shape P have?



Step 1. Corners are what this question is about, so I test each corner on its own and ignore the side lengths completely.
Step 2. Holding a card corner against each of P's four corners, all four fit exactly.
square corners on P = **4**

Try it: Shape R above has two corners that fit a card corner and two that do not. How many square corners does R have?

check: **2**

Skill 2 — Sorting a Set into Attribute Groups

One shape, many groups. A shape joins *every* group whose attribute it has, so the groups overlap — a square has four square corners *and* four equal sides *and* two pairs of parallel sides.

To sort: pick one attribute, walk through every shape, and count only the shapes that pass.

Example: A set of shapes is sorted. How many shapes have four square corners?
Step 1. The group is defined by one attribute, so I go shape by shape and ask only that one question, instead of trying to name each shape.
Step 2. Testing the corners of every shape in the set, three of them pass; the rest have corners that do not fit a card corner.
shapes with four square corners = **3**

Try it: In that same sorted set, two of the shapes have four sides of the same length. How many shapes are in the four-equal-sides group?

check: **2**

Skill 3 — Using Attribute Groups in a Real Count

Two steps, in this order: (1) decide which kinds of shape belong to the group the question names; (2) add only those counts.

Adding every number in the table answers "how many altogether," which is a different question — read what the group is before you add.

Example: A shop sold 6 square tiles, 8 rectangle tiles and 3 trapezoid tiles. How many quadrilateral tiles is that?

Step 1. Every one of the three kinds has four straight sides, so all three belong to the group named, and no tile is left out.

Step 2. Add all three counts: $6 + 8 + 3$.

quadrilateral tiles = **17**

Try it: From the same sales (6 square, 8 rectangle, 3 trapezoid), how many tiles had four square corners?

check: **14**

(!) Watch out: equal sides and square corners are different attributes. A rhombus has four equal sides but no square corners; a rectangle has four square corners but not four equal sides. Only a square has both, which is why a square lands in so many groups at once.